

Namita Chougule

DATA ENGINEER

USA | 301-284-8371 | [Email](#) | [LinkedIn](#) | [Github](#) | [Portfolio](#)

SUMMARY

Data Engineer with **4 years** of experience building and optimizing scalable data platforms, real-time streaming pipelines, and cloud-based data solutions across AWS environments. Hands-on expertise in **Apache Spark, Kafka, Snowflake, Databricks**, and **Airflow**, with a proven ability to develop **batch** and **streaming ELT pipelines**, improve query performance, and enhance production reliability. Experienced in processing high-volume datasets, optimizing multi-terabyte Spark workloads, and implementing data quality frameworks to support trusted analytics and **machine learning initiatives**. Strong foundation in distributed data systems, performance tuning, and delivering resilient, business-critical data solutions.

SKILLS

Programming Languages: Python, SQL, Bash Shell Scripting

Big Data & Distributed Processing: Apache Spark (PySpark), Apache Kafka, Delta Lake, Databricks, AWS EMR, StreamSets

Data Warehousing & Modeling: Snowflake, Dimensional Modeling (Star Schema), Data Warehouse Design, Query Optimization

Data Engineering & Pipeline Development: Batch & Streaming ELT/ETL Pipelines, Real-Time Data Processing, Data Ingestion Frameworks, Spark Performance Tuning

Workflow Orchestration & Data Quality: Apache Airflow, Great Expectations, Data Validation Frameworks, Pipeline Monitoring & Alerting

Cloud & Infrastructure: AWS (S3, EC2, EMR, VPC), Terraform, CI/CD

Core Competencies: Distributed Data Systems, Production Support & Reliability, Performance Optimization, Data Lake Architecture, Troubleshooting & Root Cause Analysis

EXPERIENCE

Quantum Integrators, USA | July 2025 – Current | Data Engineer

- Developed reusable feature pipelines and supported feature store initiatives using **Python** and **Databricks**, improving data availability for downstream **ML** workloads by **25%**.
- Improved analytical query performance by **35%** for customer and operational dashboards by **optimizing SQL** transformations and implementing multi-layer **Snowflake** data models.
- Built **batch and streaming ELT pipelines** using **StreamSets** and **Apache Kafka** to ingest and consolidate over **1.5 million** records daily from disparate on-premise systems into a centralized lakehouse environment.
- Enhanced production reliability for critical client data pipelines by implementing automated testing, monitoring, and GitOps-based deployment practices that supported **99.95% SLA** objectives.

NTT DATA Services, India | Jan 2021 - Nov 2023 | Data Engineer

- Developed real-time ingestion pipelines using Apache **Kafka** and **PySpark** Structured Streaming on **AWS EMR**, enabling the platform to process peak volumes of **15,000 events per second** and reducing data delivery latency from 6 hours to under 15 seconds.
- Contributed to the **migration** of legacy data pipelines to a **Delta Lake architecture** by implementing file optimization strategies and partitioning improvements, reducing storage utilization by **42%** and improving query efficiency.
- Enhanced batch orchestration workflows using Apache **Airflow** by developing dynamic DAGs, custom sensors, and retry mechanisms that improved pipeline reliability and supported **99.9% SLA** targets for critical reporting workloads.
- Designed **dimensional data models** and optimized **Snowflake** workloads through warehouse tuning and clustering strategies, reducing average query execution times by **68%** and lowering compute consumption by **31%**.
- Integrated **Great Expectations** into production pipelines to automate data validation checks and Slack-based alerting, improving the detection of data quality issues before downstream consumption.
- Developed **Terraform** modules to standardize provisioning of data platform resources across development, staging, and production environments, reducing deployment effort from days to under **20 minutes**.
- Optimized **multi-terabyte Spark** workloads by replacing inefficient joins with broadcast strategies, tuning executor configurations, and mitigating data skew, reducing nightly processing times by 4.5 hours and eliminating recurring OOM failures.

EDUCATION

Master of Science in Computer Science

George Mason University, USA | GPA: 3.87/4.00

May 2025

Bachelor of Engineering in Computer Science & Engineering

Visvesvaraya Technological University, India | GPA: 8.18/10.00

Aug 2020

CERTIFICATIONS

AWS Certified Data Engineer – Associate | Agile & Scrum (NLCI) | Oracle Data Engineer (NLCI) | Data Warehousing (NLCI)

PROJECT

AI-Powered Hybrid Chatbot | Python, React, GPT, Pinecone, RAG

- Built a production-ready RAG system using GPT and Pinecone for contextual, document-aware responses.
- Developed FastAPI backend and React frontend for real-time conversational interactions.
- Implemented fallback retrieval mechanisms to improve response accuracy and system robustness.